

GUJARAT UNIVERSITY

B.E. SEMESTER III (I.T.) Reg. EXAMINATION

Data Structures & Algorithms

Thursday, 20th December, 2001]

[Time : 2-30 p.m. to 5-30 p.m.

Instructions :

- (1) Answer to the *two* sections must be written in *separate* answer-books.
- (2) Figure to the right indicate full marks.
- (3) Assume suitable data, if necessary.

SECTION I

Answer the following (any six) :

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- (a) Give the difference between a data type and a data structure. Also give examples of nonprimitive data structures.
- (b) What is recursion? Discuss its advantages and disadvantages.
- (c) Do the conversions as mentioned below :
 $A + B * (C + D) / F + D * E$ (to postfix)
 $++a * bc * de$ (to infix)
- (d) Compare singly and doubly linked list.
- (e) Explain the storage structure for arrays giving example of a three dimensional array.
- (f) Show the evaluation of following postfix notation using stack :

 $abc * +de / -$

Take values of a, b, c, d and e as 2, 3, 4, 6 and 3 respectively.

- g Explain the structure of doubly linked circular list with head node. How circular queue can be implemented using this structure?
- 2 (a) What is stack? Write algorithm or program to implement push, pop and peek functions of stack using array.
- (b) Write algorithm or program to do the following in a singly linked list :
 - (i) insert a node at end
 - (ii) delete the node with value X
 - (iii) reverse the linked list.

OR

- 2 (a) Write algorithm or program to implement insert delete and display functions in a circular queue. Use array to represent circular queue.
- (b) Write algorithm or program to implement following functions in a doubly linked list.
 - (i) insert a node before the whose value is x.
 - (ii) insert a node after the node whose address is M.
 - (iii) Delete the node whose address is OLD.

- 3 (a) Give the structure of Hash table. Also write an algorithm to enter a name into hash table. 7
- (b) Answer the following : 9
- (i) What is deque ? Explain.
- (ii) Explain how stack can be implemented using singly linked list.
- (iii) Draw all distinct binary tree with three vertices.
- OR
- 3 (a) Give the properties of a hashing function. Explain any three hashing techniques with illustration. 8
- (b) Answer the following : 8
- (i) In an undirected graph with four nodes, if all nodes are connected to one another, calculate total degree of the graph.
- (ii) Give applications of stack.
- (iii) Write short note on threaded binary tree.

SECTION II

- 4 (a) Answer the following : 10
- (i) Define the following terms :
path, mixed graph, out degree.
- (ii) What is graph? How it can be represented using adjacency matrix?
- (iii) Define tree and binary search tree.
- (b) Compare selection sort and bubble sort.
- (b) Give the algorithm/program of merge sort to n numbers and explain it. 8
- 5 (a) Give the algorithm/program to insert a value into an ordered binary tree and do inorder and preorder traversal of the tree. 10
- (b) Explain how BFS and DFS are done for a graph which is represented using adjacency list. 6
- OR
- 5 (a) Give the algorithm/program to delete a node from a binary tree and to do postorder traversal of the tree. 10
- (b) Give the algorithm of Binary search. Also give trace of it. 6
- 6 (a) Give the algorithm of Quicksort and sort the following numbers using Quick sort. 10
22, 36, 6, 79, 26, 45, 75, 13, 31, 62
- (b) Explain indexed file organisation in detail. 6
- OR
- 6 (a) Explain various multiple key access file organisation. 6
- (b) Give the properties of B tree with illustration. 5
- (c) Write short note on Height balanced tree. 5